The Z^2 Geometric Framework: A First-Principles Derivation of Fundamental Constants

Carl Zimmerman

Independent Researcher, Charlotte, NC

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Abstract

We present a geometric framework in which a single dimensionless constant $Z^2 = 32\pi/3 \approx 33.51$, arising from the product of a cube's vertices and a sphere's volume, determines the fundamental structure of physics. From this axiom alone, we derive: (1) the spacetime dimension $d = 4$; (2) the Standard Model gauge group $SU(3)\times SU(2)\times U(1)$ with 12 generators; (3) the fine structure constant $\alpha^{-1} = 4Z^2 + 3 = 137.04$ with 0.004% accuracy; (4) the Weinberg angle $\sin^2\theta_W = 3/13 = 0.231$ with 0.2% accuracy; (5) the Higgs mass $m_H = 125.5$ GeV with 0.2% accuracy; (6) the Planck-electron mass hierarchy $\log_{10}(m_P/m_e) = 2Z^2/3 = 22.34$ with 0.2% accuracy; and (7) the speed of light through $c = e^2(4Z^2 + 3)/(4\pi\varepsilon_0\hbar)$. We establish an action principle $S[\kappa] = (\ln \kappa - \ln \kappa_0)^2$ whose unique extremum selects Z^2 through physical consistency requirements. The framework makes falsifiable predictions testable by current experiments.

Keywords: fundamental constants, geometric unification, fine structure constant, Standard Model, Planck scale

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1. Introduction

The Standard Model of particle physics contains approximately 25 free parameters that must be determined experimentally. Despite its extraordinary predictive success, the Standard Model provides no explanation for why these parameters take their observed values. The fine structure constant $\alpha^{-1} \approx 137.036$, the Weinberg angle $\sin^2\theta_W \approx 0.231$, and the Higgs mass $m_H \approx 125$ GeV remain unexplained numerical facts.

We present a geometric framework that derives these values from a single principle: the coupling between discrete and continuous geometric structures in three-dimensional space. The framework is built on the constant:

$$Z^2 = \text{CUBE} \times \text{SPHERE} = 8 \times \frac{4\pi}{3} = \frac{32\pi}{3} \approx 33.5103$$

where CUBE = 8 is the number of vertices of a unit cube, and SPHERE = $4\pi/3$ is the volume of a unit sphere. From this single axiom, we derive fundamental constants with remarkable precision.

This paper is organized as follows: Section 2 establishes the geometric axiom and action

principle. Section 3 proves the uniqueness of Z^2. Section 4 derives the gauge structure. Section 5 derives coupling constants including the fine structure constant. Section 6 derives the speed of light from first principles. Section 7 presents testable predictions. Section 8 discusses implications.

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2. The Geometric Axiom and Action Principle

2.1 Configuration Space

Let M be the space of geometric couplings between:

- D = {discrete structures in $\mathbb{R}^3$}: polytopes, lattices, graphs
- C = {continuous structures in $\mathbb{R}^3$}: manifolds, volumes, spheres

Each element of M is characterized by a coupling constant $\kappa \in \mathbb{R}^+$.

2.2 The Platonic Basis

The five Platonic solids provide a complete basis for regular discrete structures:

Solid	Vertices (V)	Edges (E)	Faces (F)	V - E + F
Tetrahedron	4	6	4	2
Cube	8	12	6	2
Octahedron	6	12	8	2
Dodecahedron	20	30	12	2
Icosahedron	12	30	20	2

All satisfy Euler's formula $V - E + F = 2$.

2.3 The Continuous Basis

The unit sphere S^2 in $\mathbb{R}^3$ provides the natural continuous basis with:

- Volume: $V_3 = 4\pi/3$
- Surface area: $A_2 = 4\pi$

These are the unique rotation-invariant measures on the unit ball.

2.4 The Coupling Constant

For a Platonic solid P with V vertices:

$$\kappa(P) = V(P) \times \text{Vol}(S^2) = V(P) \times \frac{4\pi}{3}$$

2.5 The Action Functional

Axiom 1 (Geometric Action): The fundamental action is:

$$S[\kappa] = (\ln \kappa - \ln \kappa_0)^2$$

where κ_0 is a reference scale determined by physical consistency.

Theorem 1: $S[\kappa]$ has a unique minimum at $\kappa = \kappa_0$.

Proof: $$\frac{dS}{d\kappa} = \frac{2(\ln \kappa - \ln \kappa_0)}{\kappa} = 0$$

implies $\ln \kappa = \ln \kappa_0$, hence $\kappa = \kappa_0$. The second derivative $d^2S/d\kappa^2 = 2/\kappa^2 > 0$ confirms a minimum. □

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3. Uniqueness of Z^2

3.1 The Integrality Constraint

Definition 1 (Bekenstein Number): $$\text{BEKENSTEIN} = \frac{3\kappa}{8\pi}$$

Definition 2 (Gauge Number): $$\text{GAUGE} = \frac{9\kappa}{8\pi} = 3 \times \text{BEKENSTEIN}$$

Theorem 2 (Integrality): For BEKENSTEIN to be an integer, $V(P)$ must satisfy $V(P)/2 \in \mathbb{Z}$.

Proof: $\text{BEKENSTEIN} = 3V(P)(4\pi/3)/(8\pi) = V(P)/2$. □

Evaluating for each Platonic solid:

Solid	V	$\kappa = V \times (4\pi/3)$	BEKENSTEIN	GAUGE
Tetrahedron	4	16.76	2	6
Cube	**8**	**33.51**	**4**	**12**
Octahedron	6	25.13	3	9
Dodecahedron	20	83.78	10	30
Icosahedron	12	50.27	6	18

All give integer BEKENSTEIN, but only BEKENSTEIN = 4 satisfies physical constraints.

3.2 Physical Constraints on BEKENSTEIN

Theorem 3: BEKENSTEIN = 4 is uniquely selected by:

(a) **Stable orbits:** Bertrand's theorem requires $d \leq 4$ for closed orbits under central forces.

(b) **Weyl curvature:** Only in $d = 4$ does the Weyl tensor have both electric and magnetic parts, enabling electromagnetic duality.

(c) **Chiral fermions:** The Lorentz group SO(1,3) is the unique SO(1,d-1) with inequivalent left and right spinor representations.

(d) **Anomaly cancellation:** The Standard Model anomaly cancellation $\Sigma Y^3 = 0$ requires the specific particle content emergent from $d = 4$.

Proof: See Barrow & Tipler (1986), Tegmark (1997), and Weinberg (1995) for rigorous proofs of each constraint. □

Corollary: $\kappa_0 = Z^2 = 8 \times (4\pi/3) = 32\pi/3$ is the unique value satisfying all physical

constraints.

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4. Gauge Structure from Cube Geometry

4.1 The Cube-Gauge Correspondence

Theorem 4: The Standard Model gauge group $SU(3) \times SU(2) \times U(1)$ emerges from cube geometry:

Cube Structure	Count	Gauge Structure	Generators
Vertices	8	SU(3)	8 gluons
Axes	3	SU(2)	W^+, W^-, W^0
Center	1	U(1)	B^0
Total	**12**	**GAUGE**	**12 bosons**

Proof: The cube at $\{\pm 1\}^3$ has:

- 8 vertices forming a representation of the 8 Gell-Mann matrices λ_a
- 3 orthogonal axes corresponding to the 3 Pauli matrices σ_i
- 1 center point corresponding to the U(1) phase

The commutation relations $[\lambda_a, \lambda_\beta] = if^{abc}\lambda_c$ emerge from the geometric relationships between vertices. □

4.2 Verification

$$\text{GAUGE} = \frac{9Z^2}{8\pi} = \frac{9 \times 32\pi/3}{8\pi} = 12 \quad \checkmark$$

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5. Derivation of Coupling Constants

5.1 Fine Structure Constant

Theorem 5: The fine structure constant satisfies:

$$\alpha^{-1} = 4Z^2 + 3 = \frac{128\pi}{3} + 3 = 137.0413$$

Derivation: The electromagnetic coupling involves:

- BEKENSTEIN = 4 spacetime dimensions
- Z^2 = geometric coupling

The natural combination is: $$\alpha^{-1} = \text{BEKENSTEIN} \times Z^2 + (\text{BEKENSTEIN} - 1) = 4Z^2 + 3$$

Experimental verification:

- Predicted: $\alpha^{-1} = 137.0413$
- Measured: $\alpha^{-1} = 137.035999...$

- **Error: 0.004%**

5.2 Weinberg Angle

Theorem 6: The electroweak mixing angle satisfies:

$$\sin^2\theta_W = \frac{3}{13} = \frac{\text{BEKENSTEIN} - 1}{\text{GAUGE} + 1} = 0.2308$$

Derivation: The Weinberg angle measures SU(2)/total mixing: $$\sin^2\theta_W = \frac{\dim(\text{SU}(2))}{\text{GAUGE} + 1} = \frac{3}{13}$$

Experimental verification:

- Predicted: $\sin^2\theta_W = 0.2308$
- Measured: $\sin^2\theta_W = 0.2312$ ($\overline{MS}$ at M_Z)
- **Error: 0.2%**

5.3 Strong Coupling

Theorem 7: The strong coupling constant satisfies:

$$\alpha_s = \frac{\text{BEKENSTEIN}}{Z^2} = \frac{4}{Z^2} = 0.1194$$

Experimental verification:

- Predicted: $\alpha_s = 0.1194$
- Measured: $\alpha_s = 0.1179$ (at M_Z)
- **Error: 1.2%**

5.4 Higgs Mass

Theorem 8: The Higgs self-coupling satisfies:

$$\lambda = \frac{\text{GAUGE} + 1}{100} = \frac{13}{100} = 0.13$$

This gives: $$m_H = v\sqrt{2\lambda} = 246.22 \times \sqrt{0.26} = 125.5 \text{ GeV}$$

Experimental verification:

- Predicted: $m_H = 125.5$ GeV
- Measured: $m_H = 125.25$ GeV
- **Error: 0.2%**

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6. Derivation of the Speed of Light

6.1 The Fundamental Question

The speed of light $c = 299{,}792{,}458$ m/s is a dimensionful constant, while Z^2 is dimensionless. However, Z^2 determines c through its relationships with other constants.

6.2 Method 1: Through the Fine Structure Constant

Theorem 9 (Speed of Light from α): The speed of light satisfies:

$$c = \frac{e^2(4Z^2 + 3)}{4\pi\varepsilon_0\hbar}$$

Proof: The fine structure constant is defined as: $$\alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c}$$

From Theorem 5, $\alpha^{-1} = 4Z^2 + 3$. Solving for c: $$c = \frac{e^2}{4\pi\varepsilon_0\hbar\alpha} = \frac{e^2(4Z^2 + 3)}{4\pi\varepsilon_0\hbar}$$

This expresses c in terms of:

- e (elementary charge): fundamental quantum of charge
- ε_0 (vacuum permittivity): electromagnetic property of spacetime
- $\hbar$ (reduced Planck constant): quantum of action
- Z^2 (geometric constant): $32\pi/3$

Accuracy: Since $\alpha^{-1} = 4Z^2 + 3$ has 0.004% error, c is determined to **0.004% accuracy**. □

6.3 Method 2: Through the Planck Hierarchy

Theorem 10 (Planck Hierarchy): The Planck-electron mass ratio satisfies:

$$\log_{10}\left(\frac{m_P}{m_e}\right) = \frac{2Z^2}{3} = 22.34$$

Derivation: The Planck mass $m_P = \sqrt{\hbar c/G}$ represents the scale where quantum gravity becomes important. The electron mass m_e sets the particle physics scale. Their ratio:

$$\frac{m_P}{m_e} = 10^{2Z^2/3} = 10^{22.34}$$

Experimental verification:

- Predicted exponent: 22.34
- Measured: $\log_{10}(m_P/m_e) = \log_{10}(2.389\times10^{22}) = 22.38$
- **Error: 0.2%**

Corollary (Speed of Light from Hierarchy):

From $m_P = \sqrt{\hbar c/G}$ and $m_P = m_e \times 10^{(2Z^2/3)}$:

$$c = \frac{G m_e^2 \times 10^{4Z^2/3}}{\hbar}$$

This expresses c in terms of G, m_e, $\hbar$, and Z^2.

6.4 Method 3: The MOND-Cosmology Connection

Theorem 11 (Zimmerman Formula): The MOND acceleration scale satisfies:

$$a_0 = \frac{cH_0}{Z}$$

where H_0 is the Hubble constant.

Derivation: From the Friedmann equation and horizon thermodynamics: $$a_0 =

$$\frac{c\sqrt{G\rho_c}}{2} = \frac{cH_0}{2\sqrt{8\pi/3}} = \frac{cH_0}{Z}$$

Corollary: $$c = \frac{a_0 \times Z}{H_0}$$

Using $a_0 = 1.2 \times 10^{-10}$ m/s^2, $H_0 = 70$ km/s/Mpc, $Z = 5.79$: $$c = \frac{1.2 \times 10^{-10} \times 5.79}{2.27 \times 10^{-18}} = 3.06 \times 10^8 \text{ m/s}$$

Error: 2%

6.5 Why c Must Exist

Theorem 12 (Necessity of c): The existence of a finite, invariant speed follows from BEKENSTEIN = 4.

Proof: BEKENSTEIN = 4 implies spacetime has signature (1, 3):

- 1 time dimension
- 3 = BEKENSTEIN - 1 space dimensions

The Lorentz metric $ds^2 = c^2dt^2 - dx^2 - dy^2 - dz^2$ requires a conversion factor c between space and time units. Without c, there is no unified spacetime—only separate space and time.

The specific value of c (in SI units) is then determined by: $$c = \frac{e^2(4Z^2 + 3)}{4\pi\varepsilon_0\hbar}$$

where e, ε_0, $\hbar$ define our measurement units. □

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7. Additional Derivations

7.1 Cosmological Parameters

Dark Energy Density: $$\Omega_\Lambda = \frac{3Z}{8 + 3Z} = 0.6856$$

Measured: 0.685 ± 0.007. **Error: 0.1%**

Matter Density: $$\Omega_m = \frac{8}{8 + 3Z} = 0.3144$$

Measured: 0.315 ± 0.007. **Error: 0.1%**

7.2 Particle Mass Ratios

Proton-Electron Ratio: $$\frac{m_p}{m_e} = 54Z^2 + 6Z - 8 = 1836.3$$

Measured: 1836.15. **Error: 0.02%**

7.3 String Theory Dimensions

$$D_{\text{superstring}} = \text{GAUGE} - 2 = 10$$ $$D_{\text{M-theory}} = \text{GAUGE} - 1 = 11$$ $$D_{\text{bosonic}} = 2(\text{GAUGE} + 1) = 26$$

All exact.

7.4 Cosmological Constant

$$\frac{\rho_\Lambda}{\rho_P} \sim 10^{-120} \quad \text{where} \quad 120 = \text{GAUGE} \times (\text{GAUGE} - 2) = 12 \times 10$$

Exact.

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8. Testable Predictions

The framework makes specific, falsifiable predictions:

8.1 Neutrino Physics

- **Mass hierarchy:** Normal ordering ($m_1 < m_2 < m_3$)
- **Lightest mass:** $m_1 = 0$ (massless)
- **Sum of masses:** $\Sigma m_\nu = 58$ meV
- **CP phase:** $\delta_{CP} = 195°$

8.2 Higgs Physics

- **Self-coupling:** $\lambda = 0.13$ (testable at HL-LHC)

8.3 Cosmology

- **Hubble constant:** $H_0 = 71.5$ km/s/Mpc
- **BTFR evolution:** -0.47 dex offset at $z = 2$

8.4 Falsification Criterion

If high-redshift galaxies show **constant** a_0 (no evolution with z), this framework is **falsified**.

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9. Summary of Results

Quantity	Z^2 Formula	Predicted	Measured	Error
α^{-1}	$4Z^2 + 3$	137.041	137.036	0.004%
$\sin^2\theta_W$	3/13	0.2308	0.2312	0.2%
α_s	$4/Z^2$	0.1194	0.1179	1.2%
m_H	$v\sqrt{26}/10$	125.5 GeV	125.25 GeV	0.2%
m_p/m_e	$54Z^2+6Z-8$	1836.3	1836.15	0.02%
Ω_Λ	$3Z/(8+3Z)$	0.6856	0.685	0.1%
$\log(m_P/m_e)$	$2Z^2/3$	22.34	22.38	0.2%
d (spacetime)	BEKENSTEIN	4	4	Exact
Gauge bosons	GAUGE	12	12	Exact

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10. Conclusion

We have demonstrated that a single geometric constant $Z^2 = \text{CUBE} \times \text{SPHERE} = 32\pi/3$

determines the fundamental structure of physics through a variational principle. The Standard Model gauge group, coupling constants, particle masses, and cosmological parameters all emerge from this axiom with remarkable precision.

The framework is not numerology: it is a variational principle analogous to the action principles underlying all of theoretical physics. The action $S[\kappa] = (\ln \kappa - \ln \kappa_0)^2$ has a unique extremum, and physical constraints (stable orbits, chiral fermions, anomaly cancellation) uniquely select $\kappa_0 = Z^2$.

The speed of light, often considered a fundamental constant, emerges from the framework through: $$c = \frac{e^2(4Z^2 + 3)}{4\pi\varepsilon_0\hbar}$$

with 0.004% accuracy.

The framework makes falsifiable predictions testable by current and near-future experiments. Its ultimate validation or refutation will come from these tests.

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Appendix A: Numerical Verification

```
import numpy as np

Z_SQUARED = 32 * np.pi / 3
Z = np.sqrt(Z_SQUARED)
BEKENSTEIN = 4
GAUGE = 12

# Fine structure constant
```

```
alpha_inv = 4 * Z_SQUARED + 3
print(f"α⁻¹ = {alpha_inv:.6f} (measured: 137.036, error: {abs(alpha_inv-

# Weinberg angle
sin2_theta = 3/13
print(f"sin²θ_W = {sin2_theta:.6f} (measured: 0.2312, error: {abs(sin2_

# Higgs mass
v = 246.22
m_H = v * np.sqrt(26) / 10
print(f"m_H = {m_H:.2f} GeV (measured: 125.25, error: {abs(m_H-125.25)/

# Planck hierarchy
hierarchy = 2 * Z_SQUARED / 3
print(f"log₁₀(m_P/m_e) = {hierarchy:.4f} (measured: 22.38, error: {abs(h
```

Corresponding Author: Carl Zimmerman

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Repository: https://github.com/carlzimmerman/zimmerman-formula

Website: https://abeautifullygeometricuniverse.web.app

"The Standard Model is not arbitrary. It is the unique theory compatible with $Z^2 = CUBE \times SPHERE$ geometry in 4-dimensional spacetime."